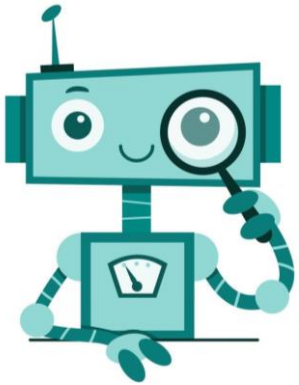
Artificial Intelligence (AI)



NCAI
Intelligent Criminology Research Lab

AGENDA

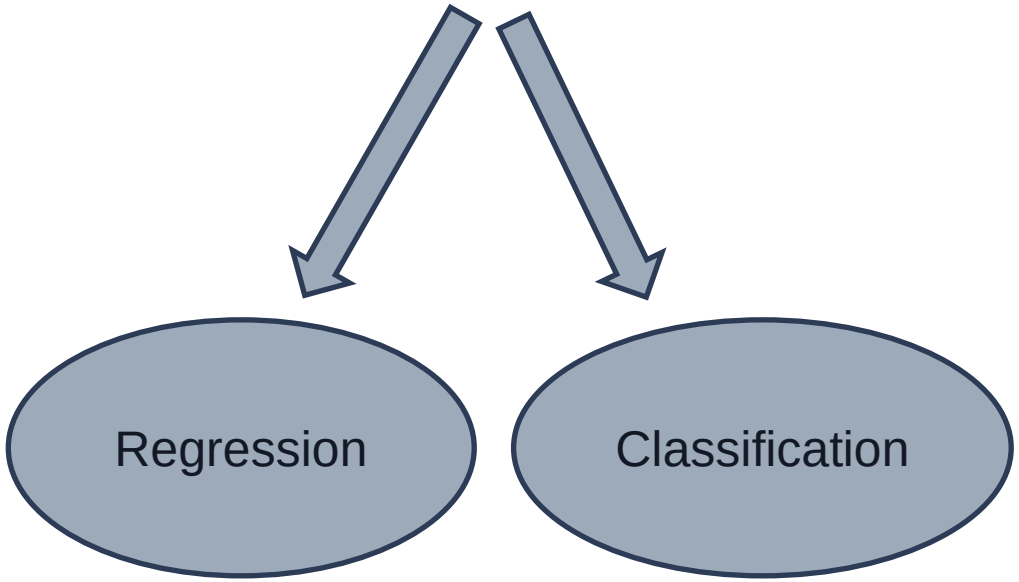
The Agenda of this presentation is:

- Types of Supervised Learning
- Regression
- Classification
- A deep look to Regression
- A deep look to Classification
- Install and import Sci-kit learn (Practical)
- Procedure



Categories

Two types of categories

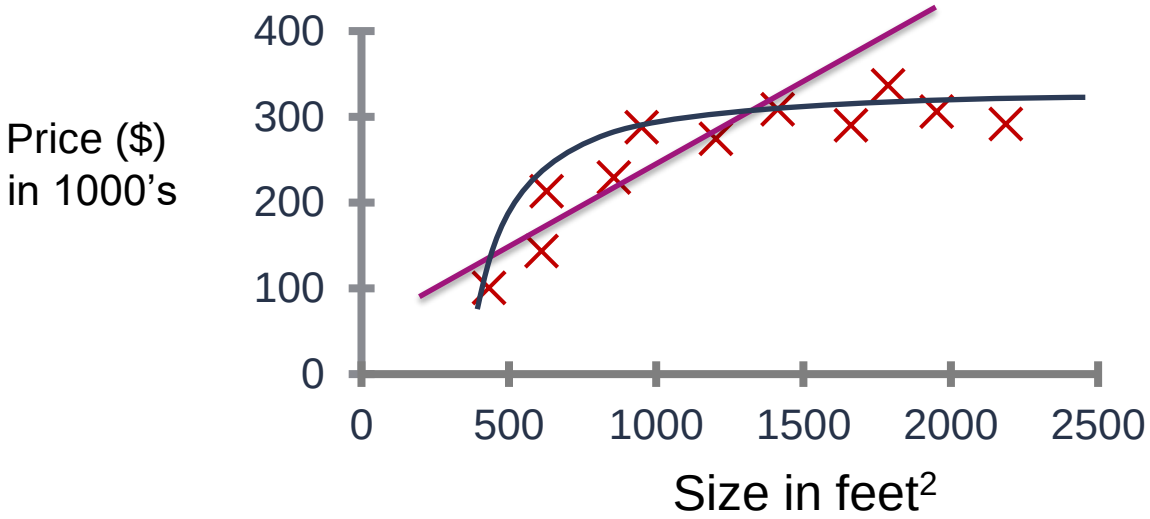


Regression

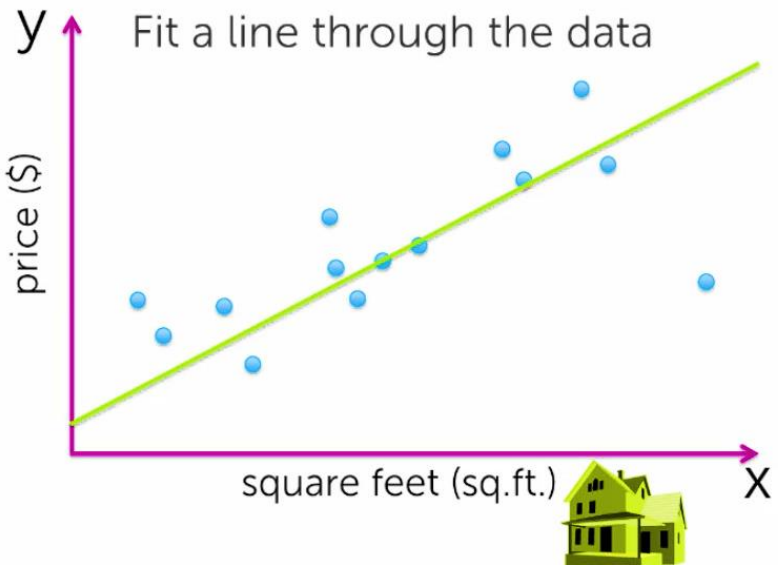
- **Regression**:
  - Predict continuous valued output.
  - predicts a single output value using training data.

Example:

Predict the house price from training data. The input variables will be price, size, locality of a house, etc.



Examples



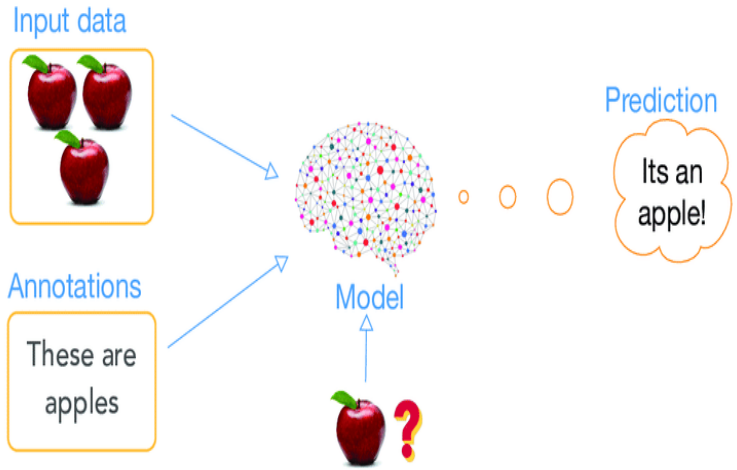
https://www.linkedin.com/

House prices prediction



https://www.researchgate.net/

Customer sentimental analysis



https://www.researchgate.net/

Predict either it is apple or not?

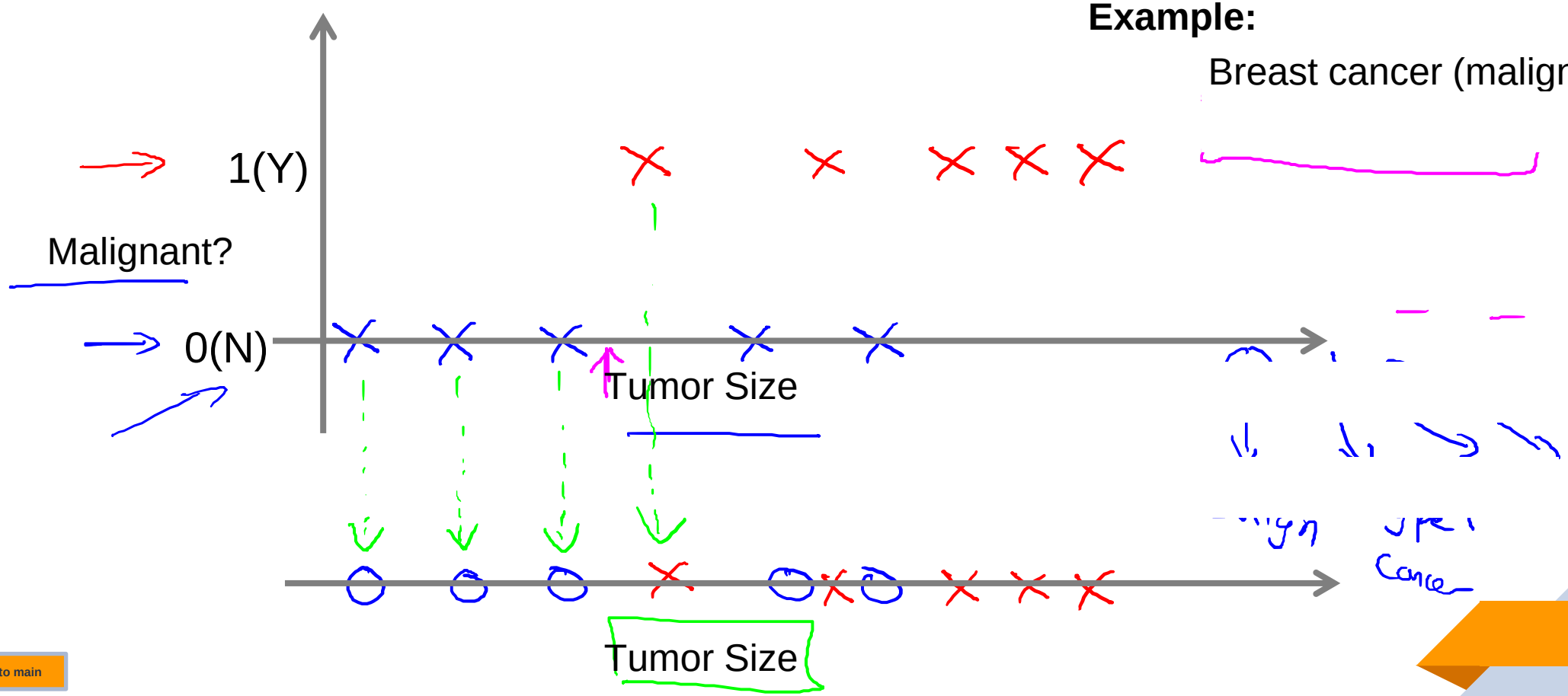
Back to main

Classification

- **Classification:**
  Predict discrete valued output (0 or 1).

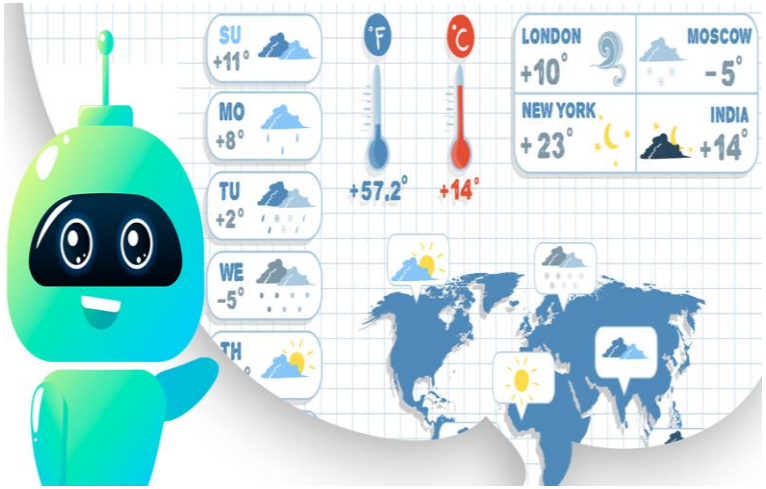
Example:

Breast cancer (malignant, benign)



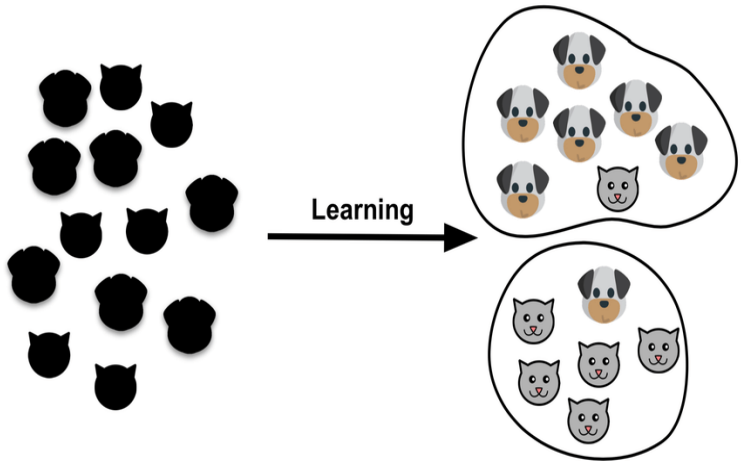
Back to main

Examples



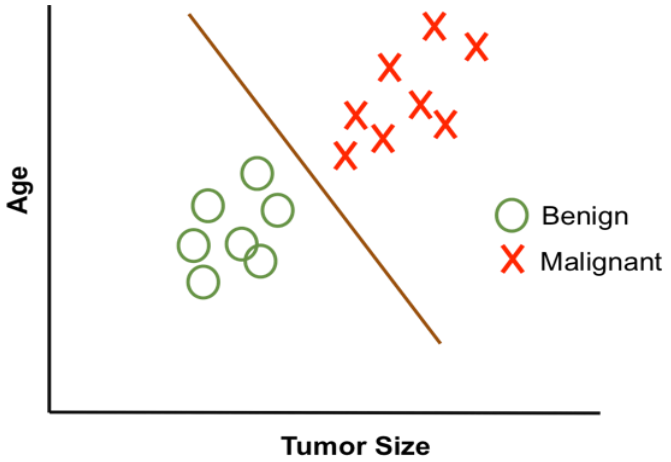
https://www.octafx.com/

How’s the weather today? prediction



https://www.researchgate.net/

Is it cat or dog prediction



https://www.ibm.com/

Breast cancer (malignant, benign)

Class Task

You're running a company, and you want to develop learning algorithms to address each of two problems.

Problem 1: You have a large inventory of identical items. You want to predict how many of these items will sell over the next 3 months.
Problem 2: You'd like software to examine individual customer accounts, and for each account decide if it has been hacked/compromised.

Should you treat these as classification or as regression problems?

- ✓ Treat both as classification problems.
- ✓ Treat problem 1 as a classification problem, problem 2 as a regression problem.
- ✓ Treat problem 1 as a regression problem, problem 2 as a classification problem.
- ✓ Treat both as regression problems.

Back to main

Regression

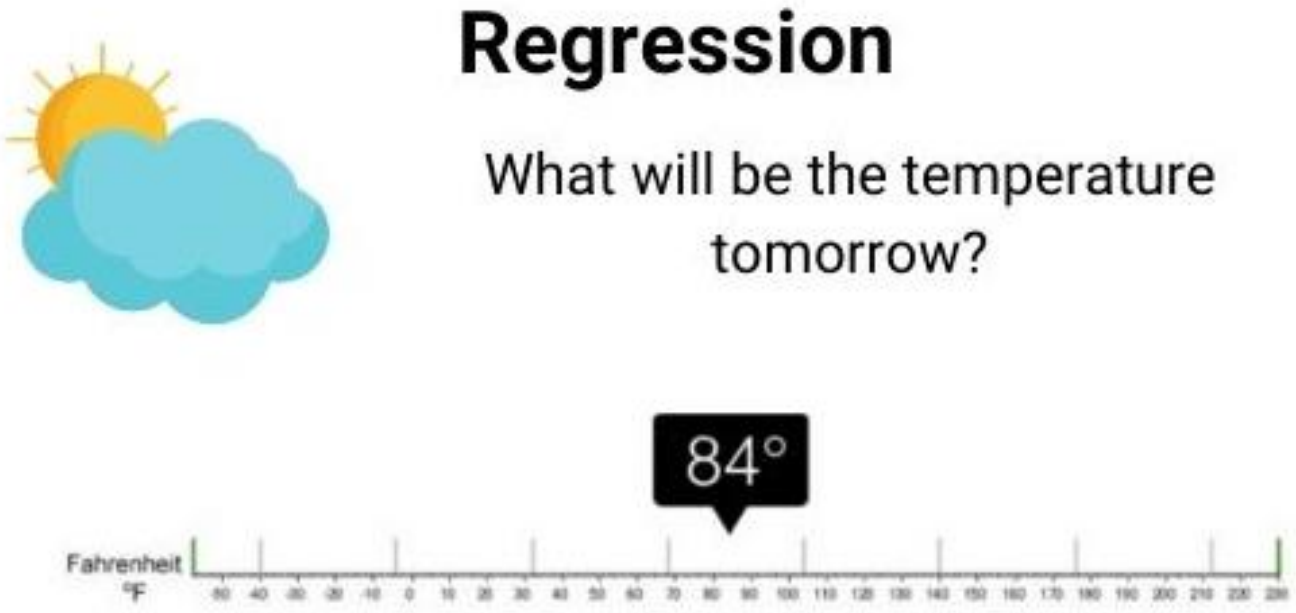
Regression Problem:

- Predict continuous valued output.
- predicts a single output value using training data.
- The output variable is a real or continuous value, such as “salary” or “weight”.
- Tries to fit data with the best hyperplane which goes through the points.

Regression Analysis:

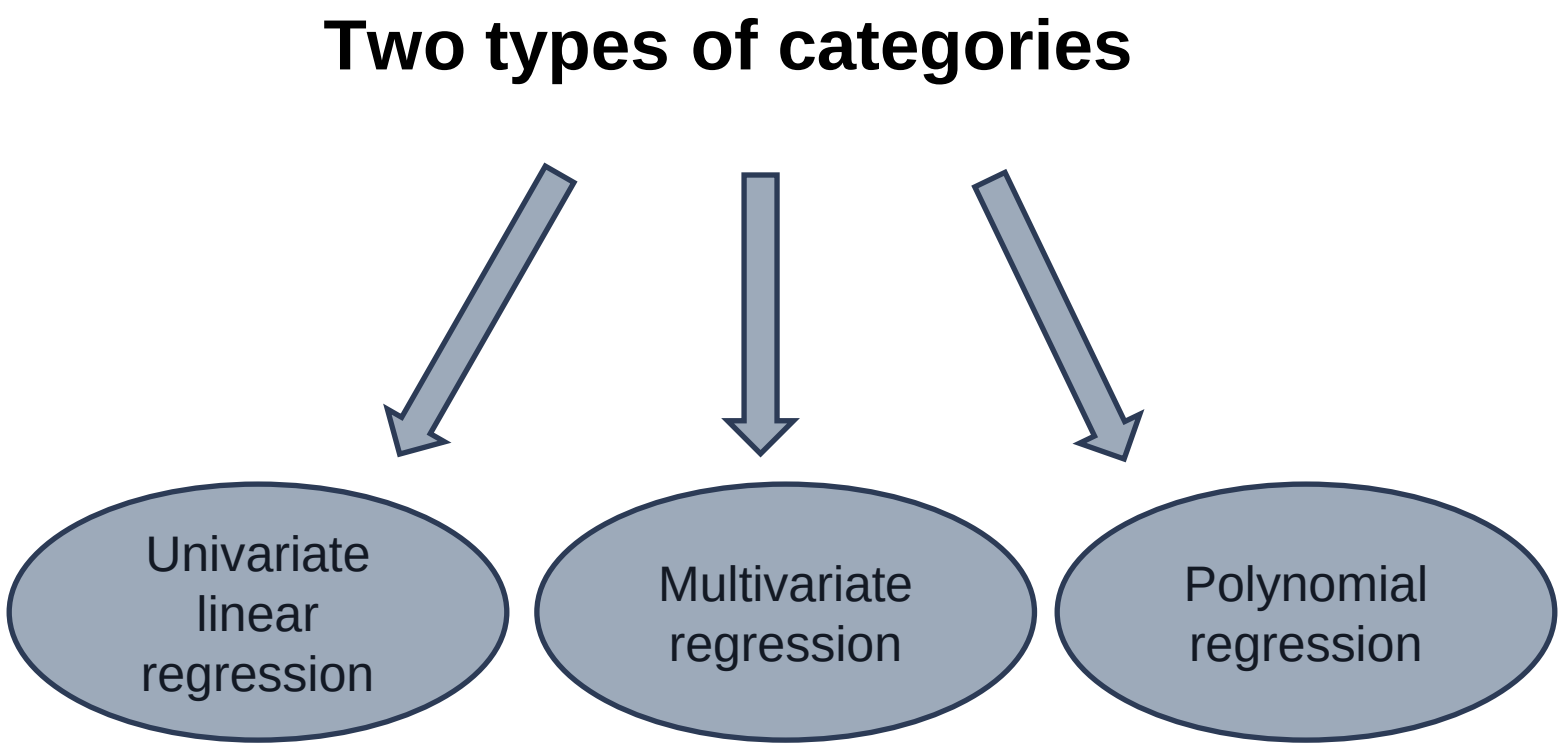
- Statistical process for estimating the relationships between the dependent variables and one or more independent variables or predictors.
- The conditional expectation of the criteria is based on predictors where the average value of the dependent variables is given when the independent variables are changed.

Example



https://www.enjoyalgorithms.com/

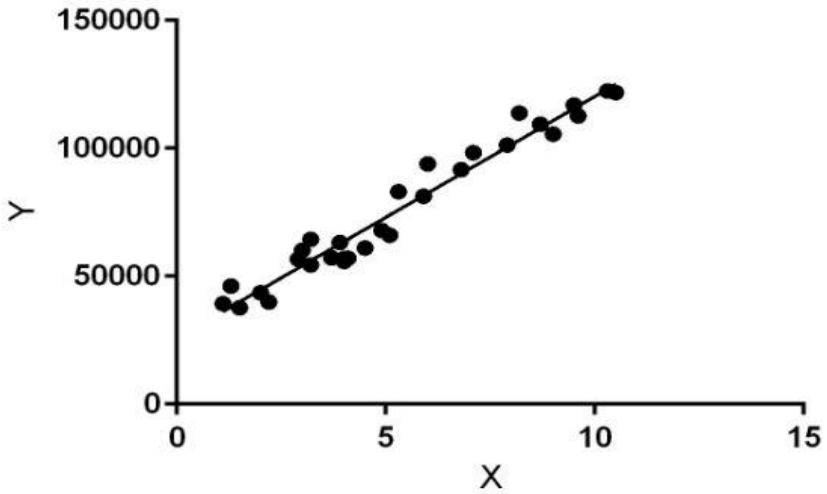
Types of Regression



Types of Regression

Linear regression:

- Machine learning algorithm based on supervised learning. Which performs a regression task.
- Regression models a target prediction value based on independent variables.
- It is mostly used for finding out the relationship between variables and forecasting.
- The formula for linear regression is: Y’ = bX + A.



https://www.geeksforgeeks.org/

Multivariate Regression

Multivariate Regression:

- One of the simplest Machine Learning Algorithm. Which comes under the class of Supervised Learning Algorithms i.e, when we are provided with training dataset.

Examples:

- A researcher has collected data on three psychological variables, four academic variables (standardized test scores), and the type of educational program the student is in for 600 high school students. She is interested in how the set of psychological variables is related to the academic variables and the type of program the student is in.
- A doctor has collected data on cholesterol, blood pressure, and weight. She also collected data on the eating habits of the subjects (e.g., how many ounces of red meat, fish, dairy products, and chocolate consumed per week). She wants to investigate the relationship between the three measures of health and eating habits.
- A property dealer wants to set housing prices which are based various factors like Size of house, No of bedrooms, Age of house, etc. We shall discuss the algorithm further using this example.

Multivariate Regression



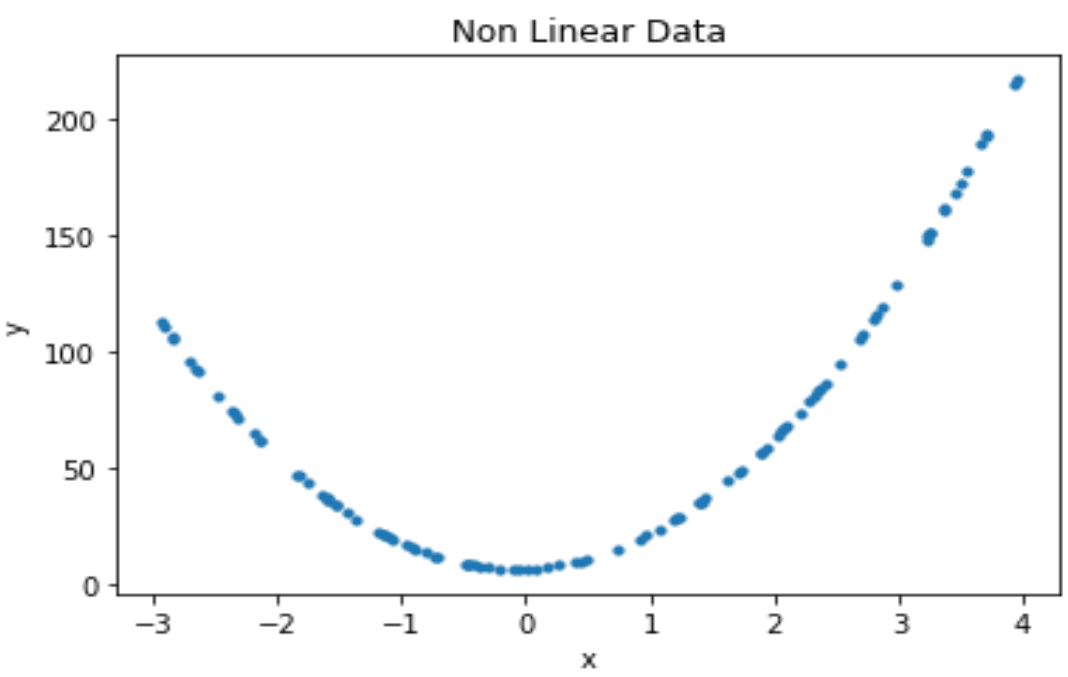
https://www.scribbr.com/

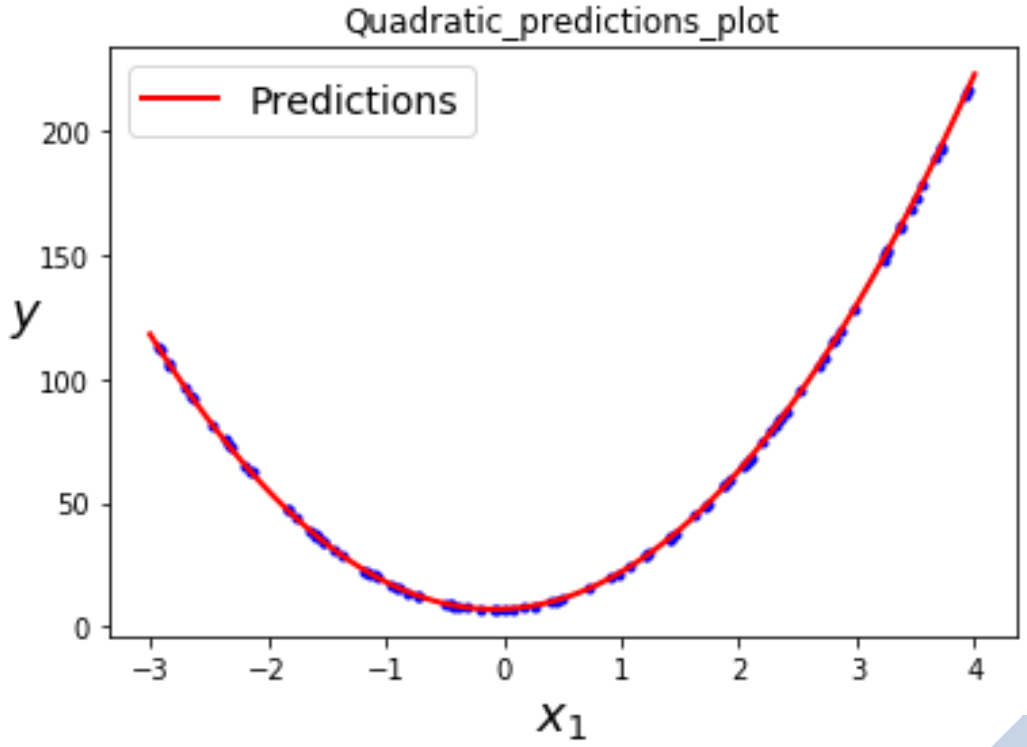
Polynomial Regression:

- Used for curvilinear data.
- Polynomial regression is fit with the method of least squares.
- The goal of regression analysis is to model the expected value of a dependent variable y in regards to the independent variable x.
- The equation for polynomial regression is:

$$\beta_0 + \beta_0 x_1 + \epsilon$$

Example


Non Linear Data
y
x
200
150
100
50
0
−3
−2
−1
0
1
2
3
4


Quadratic_predictions_plot
Predictions
y
x_1
200
150
100
50
0
−3
−2
−1
0
1
2
3
4

Types of Regression Algorithms:

- Simple Linear Regression
- Multiple Linear Regression
- Polynomial Regression
- Support Vector Regression
- Decision Tree Regression
- Random Forest Regression

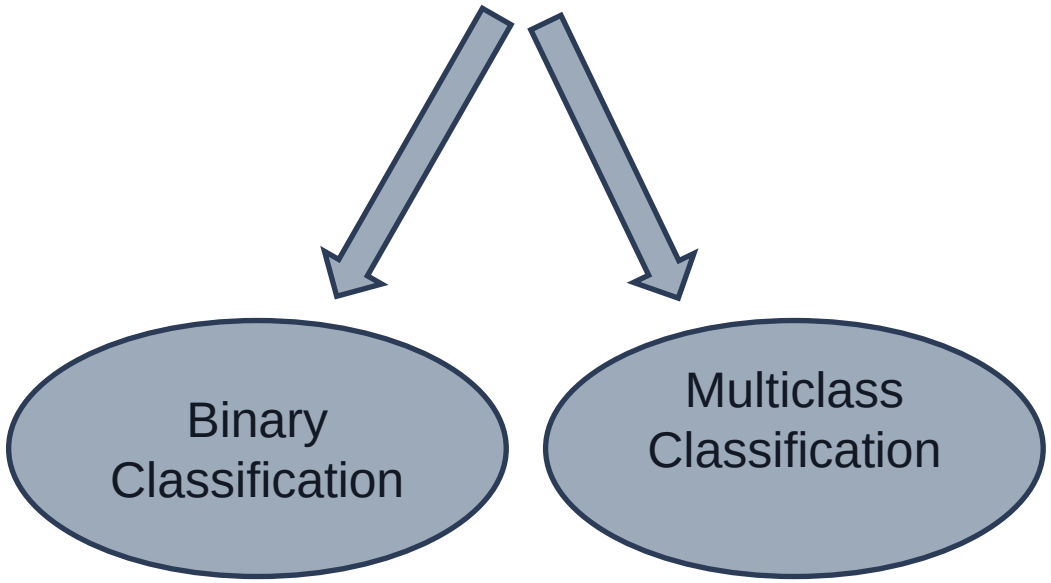
Difference between Regression and Classification

Regression Algorithm	Classification Algorithm
In Regression, the output variable must be of continuous nature or real value.	In Classification, the output variable must be a discrete value.
The task of the regression algorithm is to map the input value (x) with the continuous output variable(y).	The task of the classification algorithm is to map the input value(x) with the discrete output variable(y).
Regression Algorithms are used with continuous data.	Classification Algorithms are used with discrete data.
In Regression, we try to find the best fit line, which can predict the output more accurately.	In Classification, we try to find the decision boundary, which can divide the dataset into different classes.
Regression algorithms can be used to solve the regression problems such as Weather Prediction, House price prediction, etc.	Classification Algorithms can be used to solve classification problems such as Identification of spam emails, Speech Recognition, Identification of cancer cells, etc.
The regression Algorithm can be further divided into Linear and Non-linear Regression.	The Classification algorithms can be divided into Binary Classifier and Multi-class Classifier.

Back to main

Types of Classification

Two types of categories

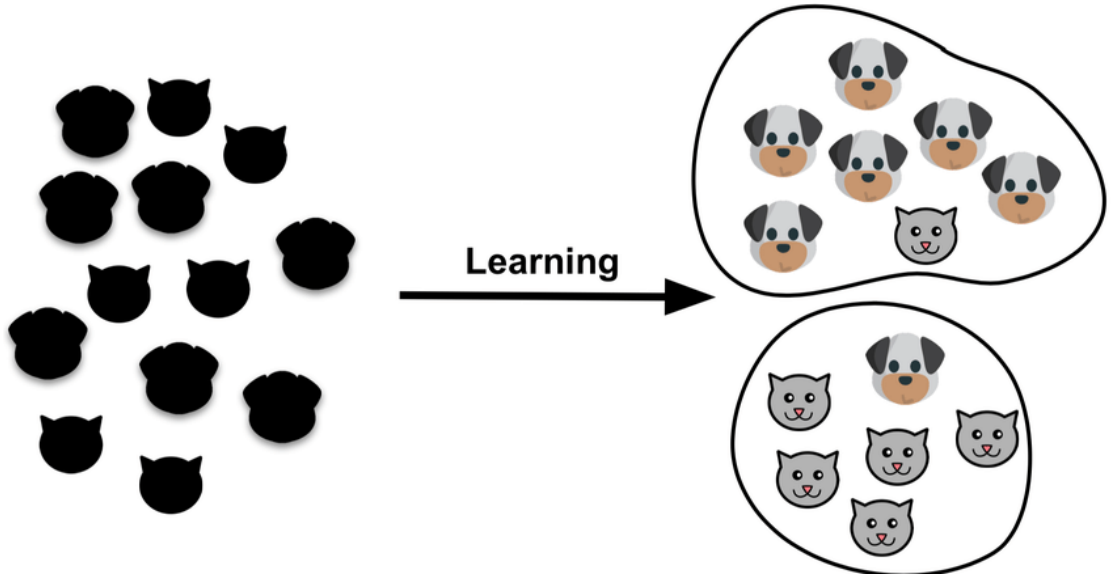


Back to main

Binary Classification

Binary Classification:

- categorize given data into 2 distinct classes.
- **Example** – On the basis of given conditions of animal classes, we have to determine whether the a class has a cats or dogs.

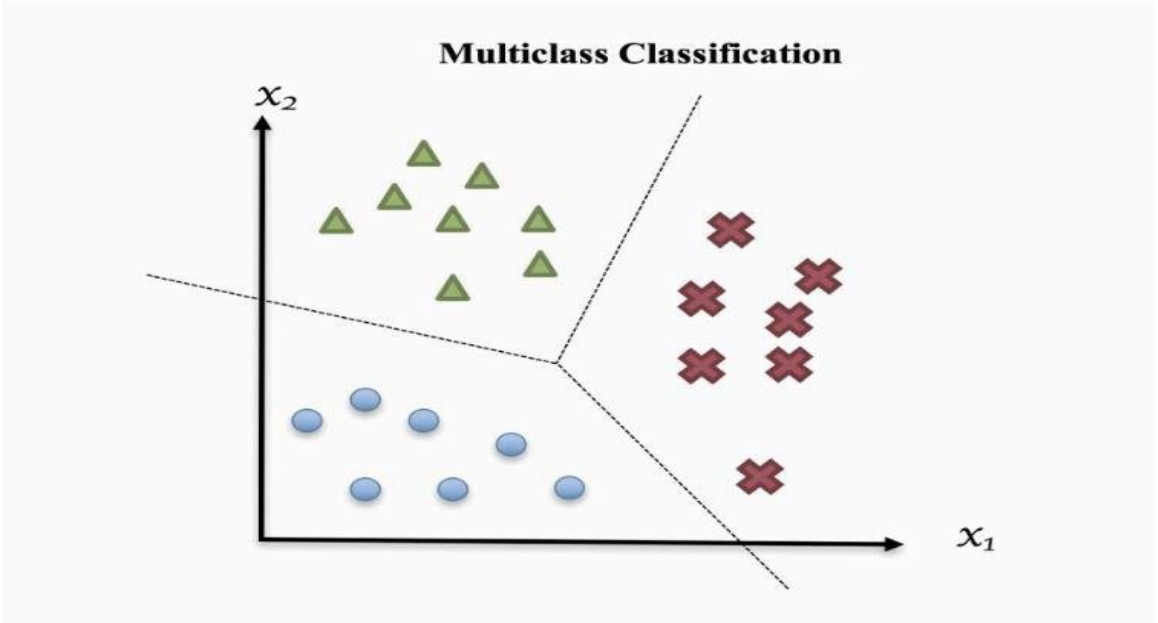


https://machine-learning-and-data-science-with-python.readthedocs.io/

Multiclass Classification

Multiclass Classification:

- The number of classes is more than 2.
- **Example** –On the basis of data about different basic shapes, we have to determine which shape does belongs to class.



https://towardsdatascience.com/

How does classification works?

Suppose we have to predict whether a given patient has a certain disease or not, on the basis of 3 variables, called features.

This means there are two possible outcomes:

- The patient has the said disease. Basically, a result labeled “Yes” or “True”.
- The patient is disease-free. A result labeled “No” or “False”.
- This is a binary classification problem.

- set of observations called **training data set**, which comprises sample data with actual classification results.
- train a model, called **Classifier** on this data set, and use that model to predict whether a certain patient will have the disease or not.

The outcome, thus now depends upon :

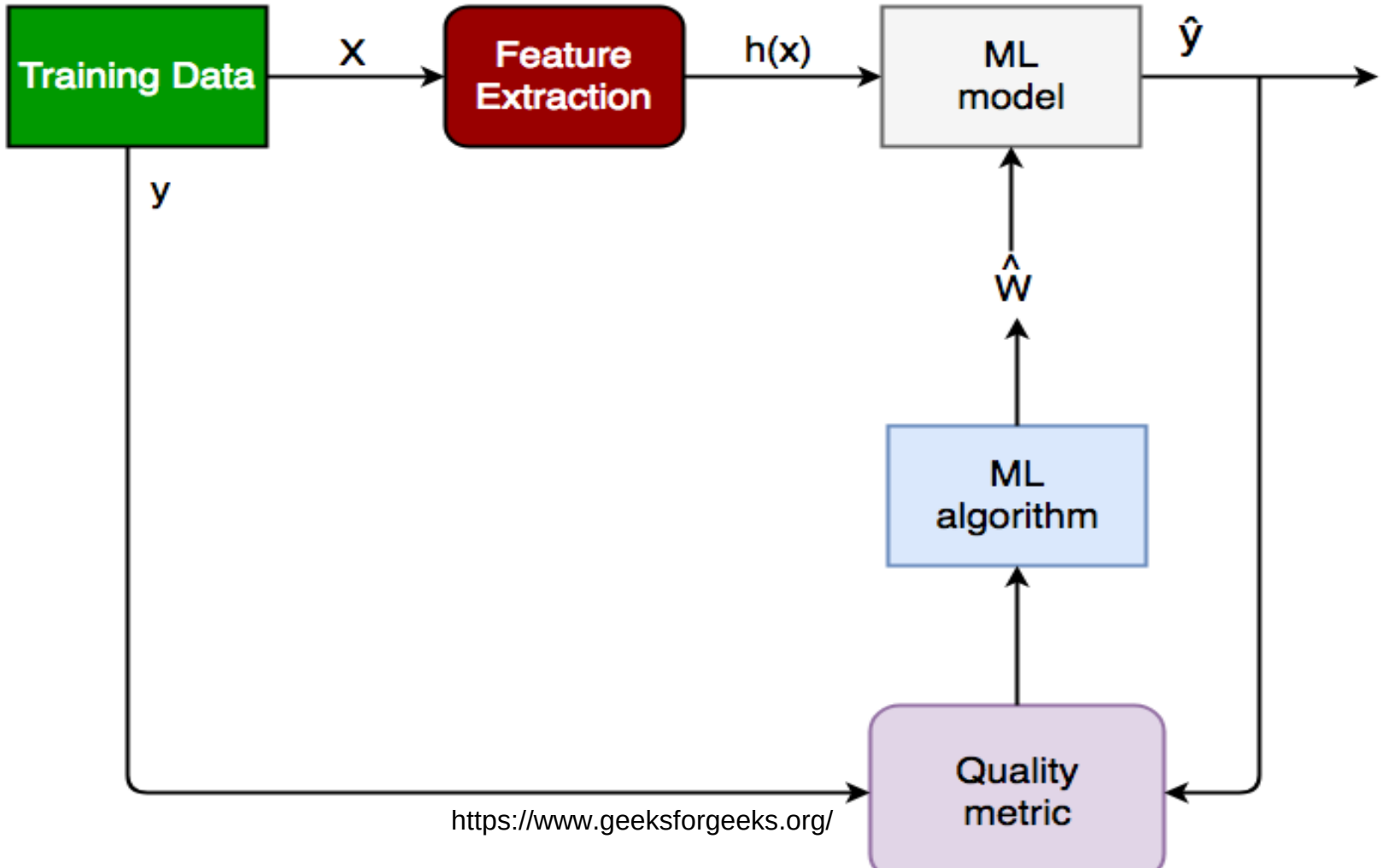
How well these features are able to “map” to the outcome.

The quality of our data set.

How well our Classifier generalizes this relationship between the features and the outcome.

The values of the x1 and x2.

Generalized Classification Block Diagram.



https://www.geeksforgeeks.org/

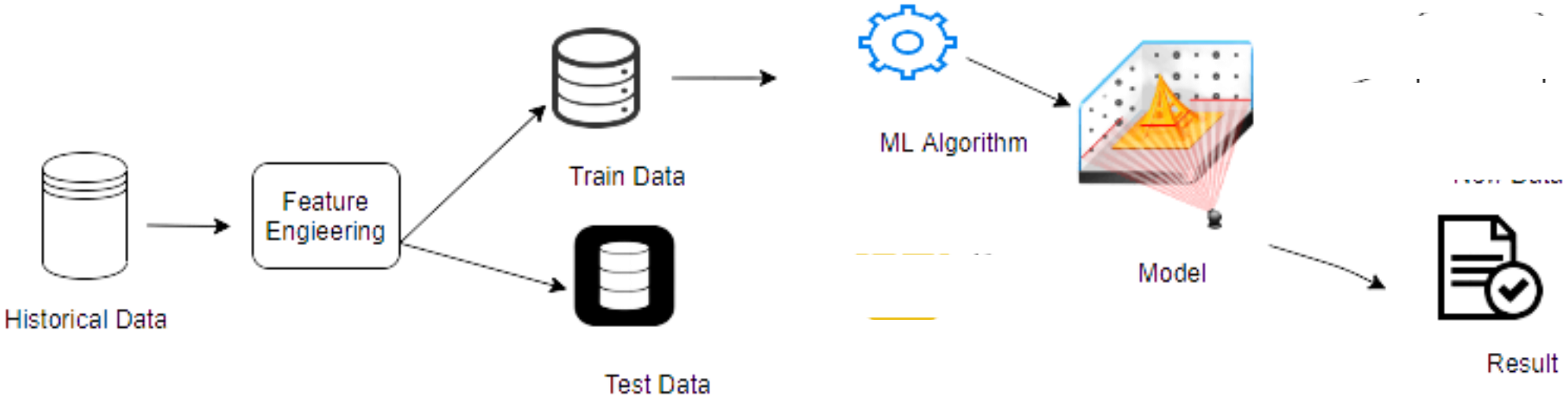
Types of Classifiers (algorithms)

There are various types of classifiers. Some of them are:

- Linear Classifiers: Logistic Regression
- Tree-Based Classifiers: Decision Tree Classifier
- Support Vector Machines
- Artificial Neural Networks
- Bayesian Regression
- Gaussian Naive Bayes Classifiers

Ensemble Methods: Random Forests, AdaBoost, Bagging Classifier, Voting Classifier, Extra Trees Classifier

Procedure



https://medium.com/

Back to main

Start With a Data Set

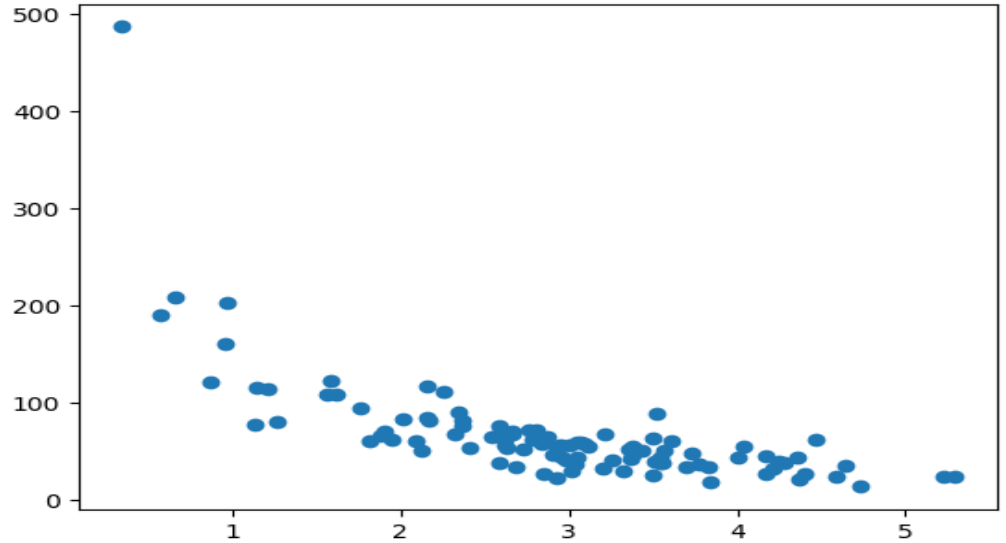
- Start with a data set you want to test.
- Our data set illustrates 100 customers in a shop, and their shopping habits.

Example

```
import numpy
import matplotlib.pyplot as plt
numpy.random.seed(2)

x = numpy.random.normal(3, 1, 100)
y = numpy.random.normal(150, 40, 100) / x

plt.scatter(x, y)
plt.show()
```



Result:

- The x axis represents the number of minutes before making a purchase.
- The y axis represents the amount of money spent on the purchase.

Split Into Train/Test

The **training set** should be a random selection of **80%** of the original data.

The **testing set** should be the remaining **20%.**

Syntax:

```
train_x = x[:80]
train_y = y[:80]

test_x = x[80:]
test_y = y[80:]
```

Train-Test Split:

- The train-test split is a technique for evaluating the performance of a machine learning algorithm.
- It can be used for classification or regression problems and can be used for any supervised learning algorithm.

Procedure:

- The **procedure** involves taking a dataset and dividing it into two subsets.
- The first subset is used to fit the model and is referred to as the **training dataset.**
- The second subset is not used to train the model; instead, the input element of the dataset is provided to the model, then predictions are made and compared to the expected values. This second dataset is referred to as the **test dataset.**

- **Train Dataset**: Used to fit the machine learning model. **80%** for training.
- **Test Dataset:** Used to evaluate the fit machine learning model. **20%** for testing.

The **objective** is to estimate the performance of the machine learning model on new data.
The train-test procedure is **appropriate** when there is a **sufficiently large dataset** available.

Display the Training Set

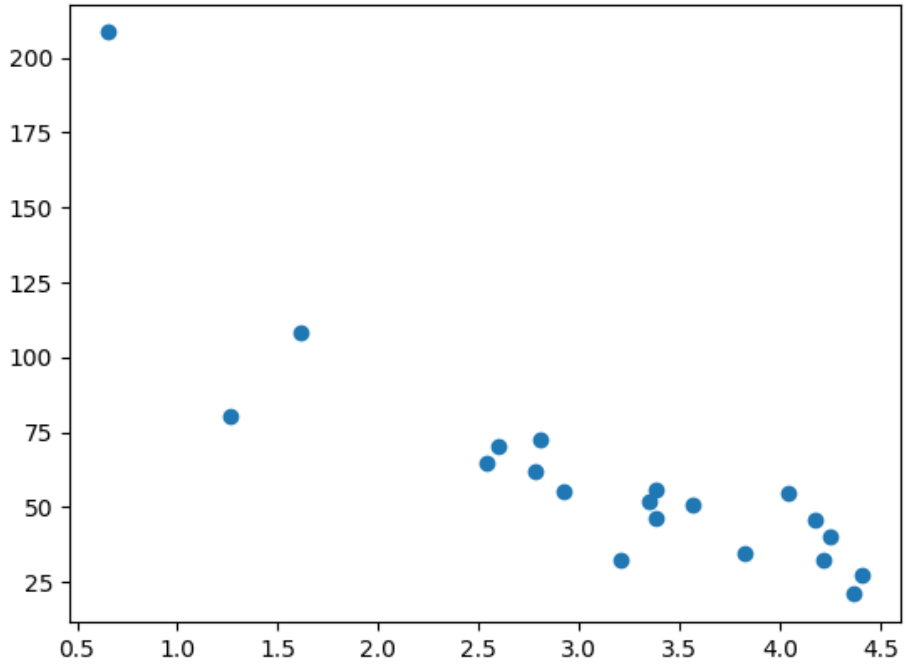
To make sure the testing set is not completely different, we will take a look at the testing set as well.

Example

```
plt.scatter(test_x, test_y)
plt.show()
```

Result:

- ✓ The testing set also looks like the original data set:



Fit the Data Set

- Best fit would be a polynomial regression, so let us draw a line of polynomial regression.
- To draw a line through the data points, we use the plot() method of the matplotlib module:

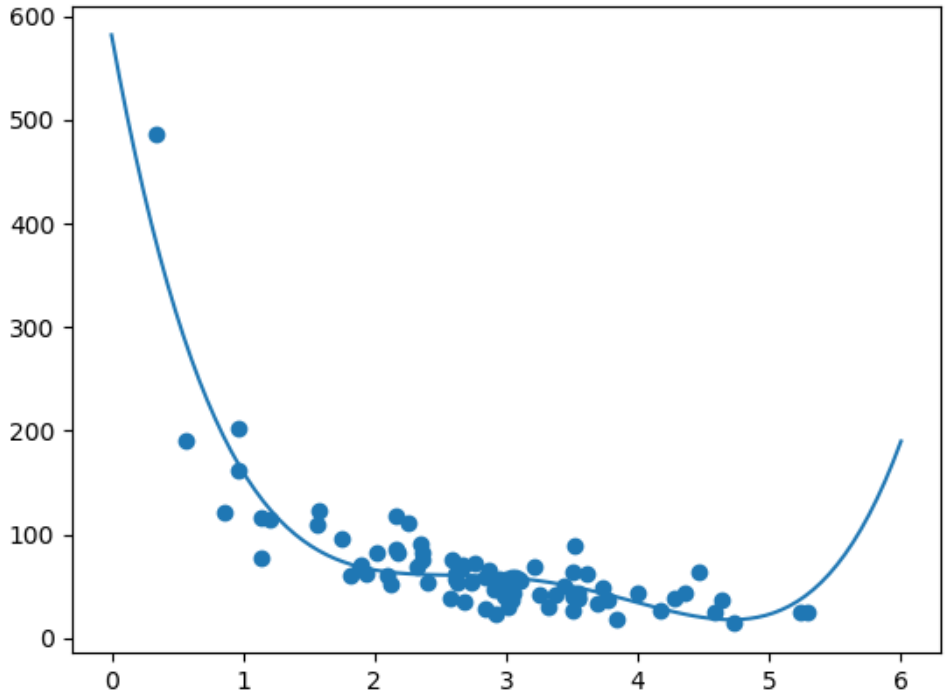
Example:

Draw a polynomial regression line through the data points:

```
import numpy
import matplotlib.pyplot as plt
numpy.random.seed(2)
x = numpy.random.normal(3, 1, 100)
y = numpy.random.normal(150, 40, 100) / x

 train_x = x[:80]          test_x = x[80:]
   train_y = y[:80]          test_y = y[80:]

mymodel = numpy.poly1d(numpy.polyfit(train_x, train_y, 4))
myline = numpy.linspace(0, 6, 100)
plt.scatter(train_x, train_y)
plt.plot(myline, mymodel(myline))
plt.show()
```



Predict Values

✓ Now that we have established that our model is **OK**, we can start predicting new values.

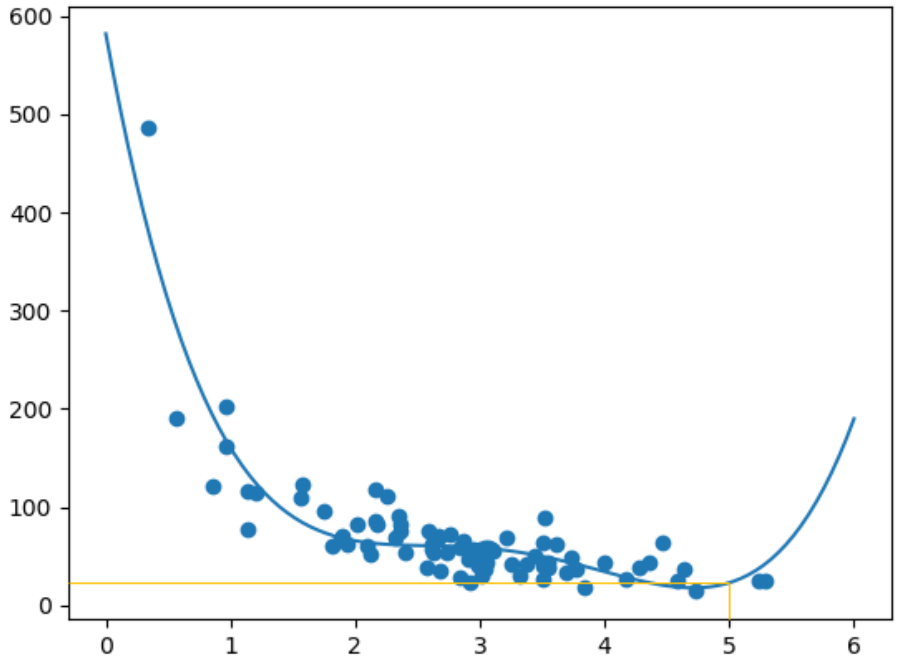
Example

How much money will a buying customer spend, if she or he stays in the shop for 5 minutes?

```
print(mymodel(5))
```

Result:

The example predicted the customer to spend 22.88 dollars, as seems to correspond to the diagram:



Back to main